TEAM ELEVATE TECH | A3-PS003-TC068

AIORI-3 HACKATHON



FLOWPULSE-FL

COMMUNICATION-AWARE FEDERATED TRAFFIC-FLOW FORECASTING

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WHAT IS FLOWPULSE-FL?

- FlowPulse-FL is a historical traffic-volume forecasting simulator using federated learning.
- Four simulated clients use partitions of one public dataset. In a single-process simulation, each client trains on its assigned data and passes model updates to a simulated central server—not raw records.
- The server combines selected client updates using sample-count-weighted FedAvg. The simulator compares Full, Random, and FlowPulse selection to explore forecasting accuracy versus simulated model-upload/download payload.

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PRESENTATION ROADMAP

Problem & Motivation

FL Background

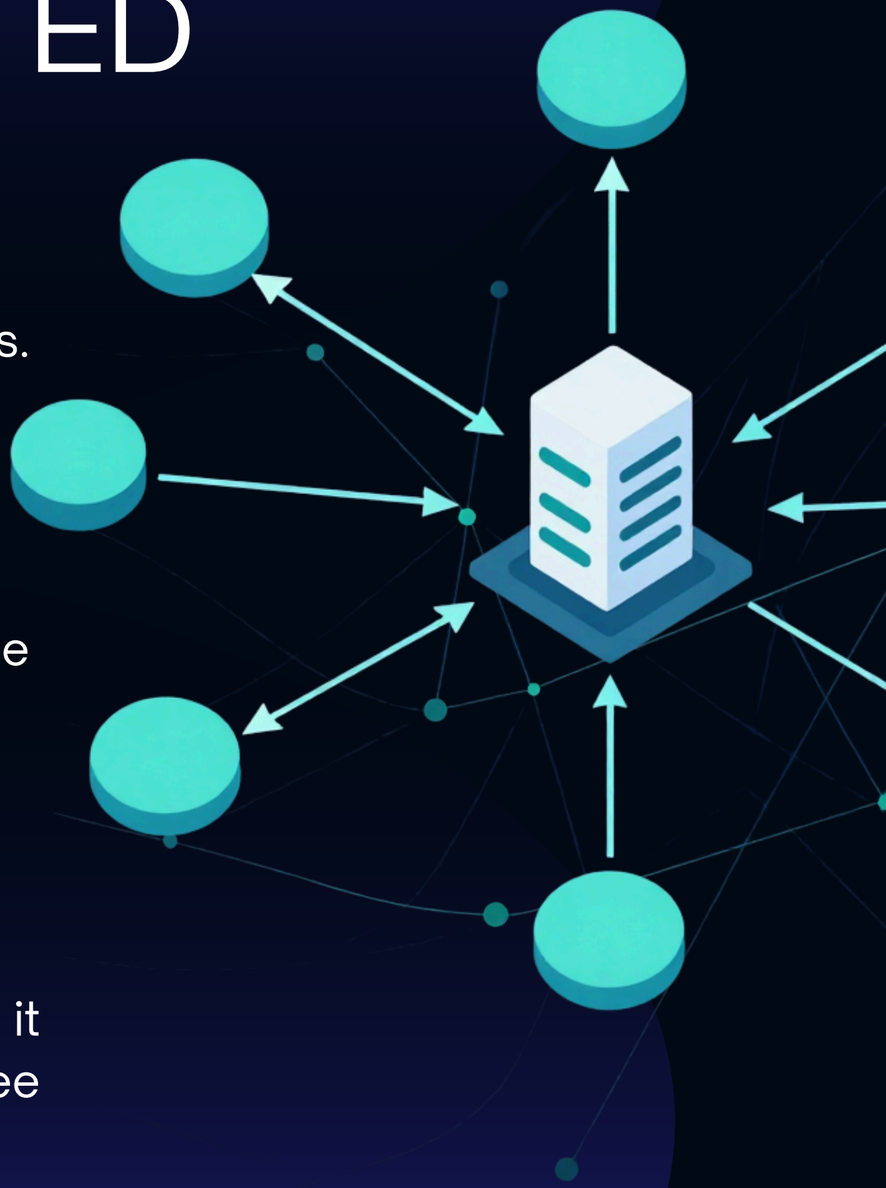
FlowPulse-FL approach

Architecture, data flow, and verified three-seed evaluation

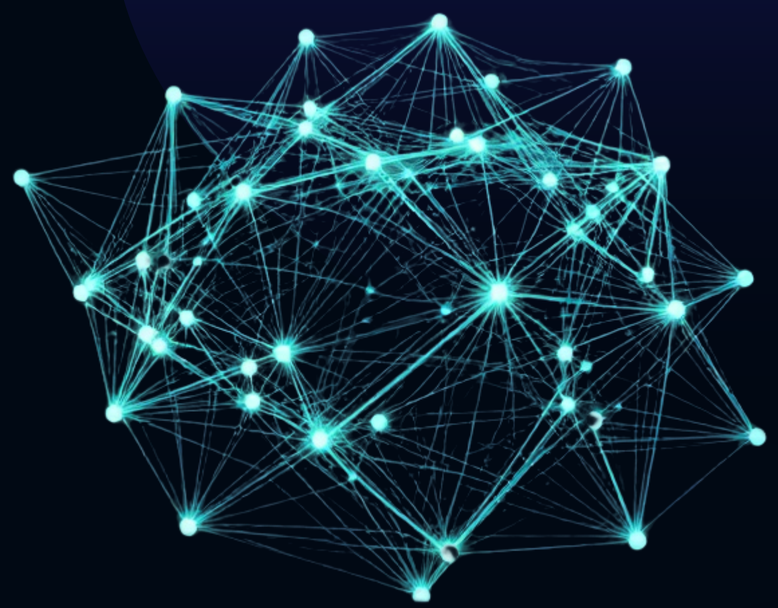
From traffic-forecasting challenges to a historical-data simulator—this deck explains our federated-learning approach, client-selection strategies, verified results, and limitations.

THE PROBLEM: DISTRIBUTED TRAFFIC DATA

- Traffic data may be collected across separate locations, where centralizing raw records can raise privacy and data-governance concerns.
- Federated learning offers an alternative: clients train on local data and share model updates for aggregation.
- Having every client exchange model updates in every round increases the model-payload cost.
- FlowPulse-FL explores this trade-off through four simulated clients partitioned from one historical dataset.
- The prototype compares forecasting accuracy and model-payload cost; it does not demonstrate physical distributed deployment or guarantee privacy.



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- FlowPulse-FL simulates four federated clients using partitions of one historical traffic-volume dataset.
- Each client trains a compact linear model to predict the next hour’s traffic volume from 24 hourly inputs.
- Within the single-process simulation, selected clients pass model updates to a simulated server, which combines them using sample-count-weighted FedAvg.
- We compare Full participation, Random selection, and FlowPulse selection to study forecasting accuracy versus model-upload/download payload.

WHY FEDERATED TRAFFIC FORECASTING?

LITERATURE & RFC REVIEW

- Federated learning enables multiple clients to train a shared model without centralizing their raw data.
- FedAvg aggregates client model updates, commonly weighting each update by the client’s number of training samples.
- Client-selection research studies the trade-off between model accuracy, communication cost, and client participation.
- Traffic-flow forecasting can use historical sensor patterns. FlowPulse-FL uses a compact linear forecasting model to keep model-update payloads small.
- This prototype is informed by federated-learning and client-selection research. It does not implement a specific production RFC, differential-privacy mechanism, or secure-aggregation protocol.





HOW FLOWPULSE-FL WORKS

- The simulated server starts each round with a shared global forecasting model.
- Full participation selects all four clients. Random and FlowPulse select clients within the configured participation budget.
- Selected clients receive the global model, train on their assigned data, and upload their updated model parameters.
- The server combines these parameters using sample-count-weighted FedAvg, then begins the next round with the updated global model.

SOLUTION OPTIMIZATION

Scenario	Clients selected per round	Model-payload bytes over 12 rounds	Result
Before: Full participation	4 of 4	19,200 bytes	Every client participates in every round
After: Budgeted selection	2 of 4	9,600 bytes	50% model-payload reduction
Random selection	2 of 4	9,600 bytes	Same payload reduction; random client ordering
FlowPulse selection	2 of 4	9,600 bytes	Same payload reduction; timeout and readiness-based selection priority

Model payload only; excludes metadata and network overhead. Accuracy differences across selection strategies were not statistically significant.

SYSTEM ARCHITECTURE SERVER–CLIENT WORKFLOW



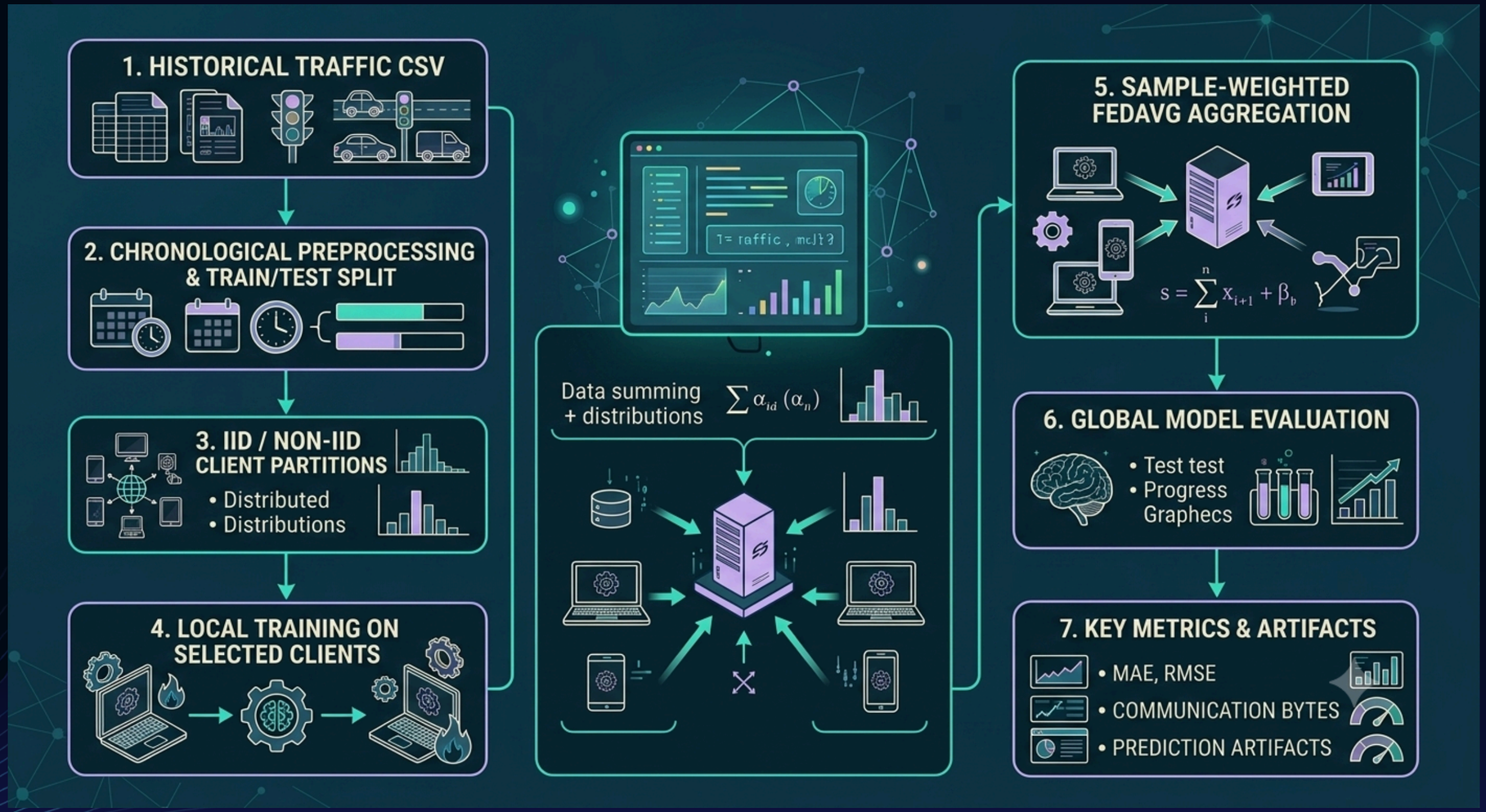
CLIENT NODES

- Four clients are simulated partitions of one historical dataset.
- Selected clients receive the global model, train locally, and upload updated parameters.
- All clients run in one process—not on physical edge devices.

FEDERATED SERVER

- Selects clients using Full, Random, or FlowPulse.
- Combines updates using sample-count-weighted FedAvg.
- Counts model uploads/downloads only—not metadata or network overhead.

DATA FLOW DIAGRAM



TECHNICAL CONTRIBUTIONS

Federated Learning

- Four simulated clients train on partitions of one historical dataset.
- A 24-input linear model forecasts next-hour traffic volume. Selected updates are combined using sample-count-weighted FedAvg.

Communication-Aware Selection

- Compares Full, Random, and FlowPulse selection.
- Over 12 rounds, both Random and FlowPulse select 2 of 4 clients per round: 9,600 bytes versus Full’s 19,200—a 50% model-payload reduction.
- Excludes metadata and network overhead.

VERIFIED THREE-SEED RESULTS

Both FlowPulse and Random use 9,600 model-payload bytes versus Full's 19,200—a 50% reduction over 12 rounds at budget 2.

Three-seed results (42, 7, 2026) show similar mean MAE/RMSE across strategies in IID and non-IID settings, without establishing statistically significant superiority.



Results | Communication | Convergence

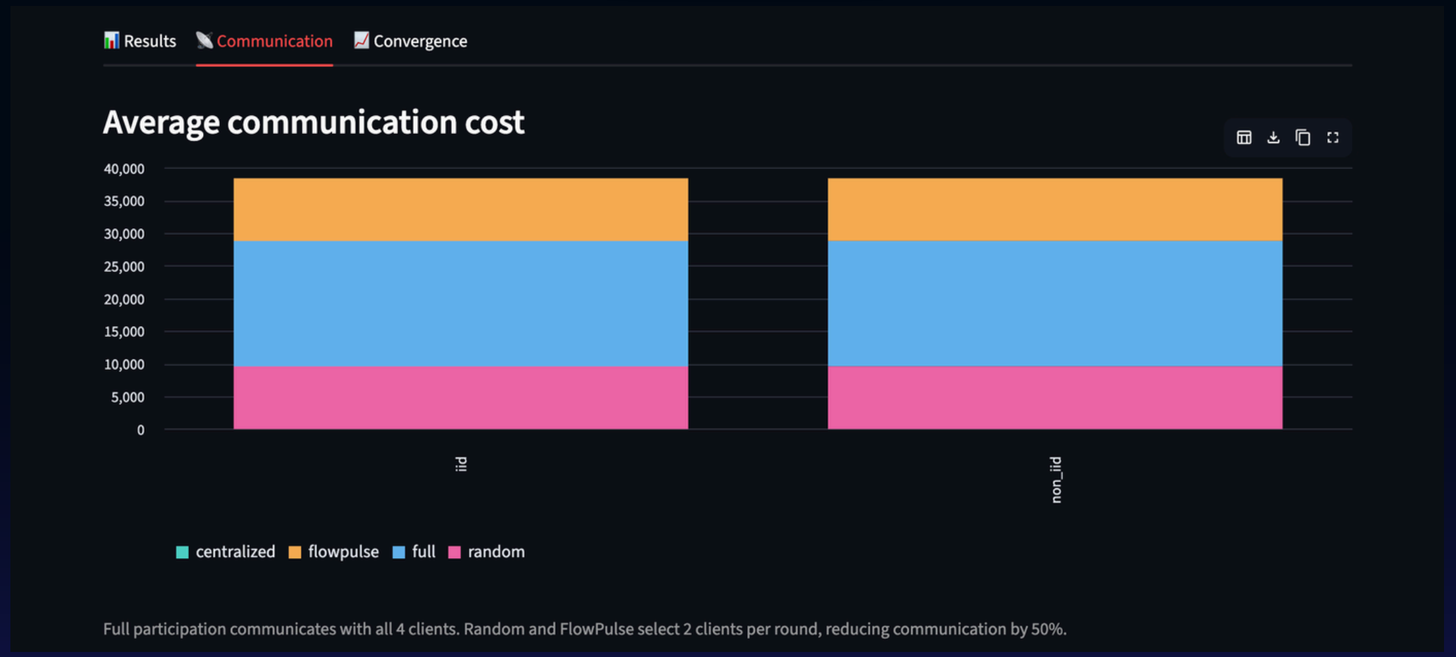
Three-seed forecasting performance

partition_mode	selection_strategy	mae_mean	mae_std	rmse_mean	rmse_std	communication_bytes_mean	communication_bytes_std	runs
iid	centralized	759.1505	10.5023	982.4509	14.7242	0	0	3
iid	flowpulse	759.346	10.2767	982.6267	14.6129	9600	0	3
iid	full	759.1539	10.5006	982.4579	14.7241	19200	0	3
iid	random	758.9825	10.6157	982.4416	14.8412	9600	0	3
non_iid	centralized	759.1505	10.5023	982.4509	14.7242	0	0	3
non_iid	flowpulse	762.66	10.4916	985.2234	13.6228	9600	0	3
non_iid	full	759.5167	10.5128	982.931	14.7148	19200	0	3
non_iid	random	756.3515	8.1718	981.0421	13.7077	9600	0	3

Mean and standard deviation across seeds 42, 7, and 2026.

COMMUNICATION REDUCTION

12 rounds · Model payload only
Full: 19,200 bytes
Random / FlowPulse: 9,600 bytes (50% less)
Excludes metadata and network overhead.



DEPLOYMENT TIMELINE & BIBLIOGRAPHY

Deployment Timeline:

- Data acquisition and hourly preprocessing
- Federated simulation and client-selection implementation
- Dashboard, tests, and benchmark-artifact generation
- Three-seed evaluation and result reconciliation
- Render deployment and public demo verification
- Final repository and presentation review

Bibliography:

- B. McMahan et al., “Communication-Efficient Learning of Deep Networks from Decentralized Data,” AISTATS, 2017.
- UCI Machine Learning Repository, “Metro Interstate Traffic Volume.”
- Streamlit Documentation, https://docs.streamlit.io/.

CONCLUSION & THANK YOU

FlowPulse-FL explores forecasting accuracy versus model payload using four simulated clients from one historical dataset.

Over 12 rounds, both Random and FlowPulse halve model payload: 9,600 vs 19,200 bytes for Full.

Three-seed results show similar mean errors, not proven superiority.

Live demo: https://a3-ps003-tc068.onrender.com

Thank you!